

Diagonalizing $A = S\Lambda S^{-1}$ via deflation

1 Problem Statement

Given a linear transformation matrix A . We want to know:

- Is A diagonalizable as $S\Lambda S^{-1}$?
- If so, how to find S matrix and Λ matrix which contain only eigenvalues on the diagonal?

2 Core idea of deflation

Assuming A is diagonalizable, we can't just find all the eigenvalues directly, and find the eigenbasis of each eigenvalue and assemble these bases to S . The assembled S is still invertible as all these bases are linearly independent. The problem lies in how to find all the eigenvalues.

Say we try to solve $\det(A - \lambda I) = 0$ as a polynomial of λ . The problem is that when the polynomial is above 5th degree, it was proven not solvable.

That is why we need to use power iteration method and deflation to keep finding all the eigenvalues one by one.

The core idea of deflation is simple, just like SVD decomposition, we can view A as sums of outer products of eigen:

$$A = \sum_{i=1}^k \lambda_i S_i Q_i^T$$

Here, S_i is the matrix as λ_i eigenspace basis. Q_i^T can be deemed as S_i^+ . Thus $Q_i^T S_i = I_{p_i \times p_i}$. Here p_i is the dim of E_{λ_i} .

Once we use power iteration method to find dominant λ_1 , we can use RREF to find basis of $(A - \lambda_1 I)$ null space. The basis is S_1 .

We can use left inverse of S_1 , aka S_1^+ , aka $(S_1^T S_1)^{-1} S_1^T$ as Q_1^T . Then we calculate:

$$A_{\text{deflated}} = A - \lambda_1 S_1 Q_1^T$$

With A_{deflated} , we can keep going to find and subtract next outer product.

Note, here, for each λ_i , we find the S_i . If we assemble $S = (S_1 \dots S_k)$, the columns of S are linearly independent. The idea is simple, because we can view the input space $\mathbb{R}^n$ as:

$$\mathbb{R}^n = E_{\lambda_1} \oplus E_{\lambda_2} \oplus \dots \oplus E_{\lambda_k}$$

The k eigenspaces are kind of like linearly independent vectors in the sense any vector $\vec{v}$ in $\mathbb{R}^n$ has only one unique decomposition in these k eigenspaces. Since inside the eigenspace E_{λ_i} , S_i matrix also contains basis as its columns. It is easy to imagine two steps process to prove S columns are linearly independent.

By now we have S and Λ . However, one thing to note is that the Q_i^T can't be assembled as $Q^T = (Q_1^T \ Q_2^T \ \dots \ Q_k^T)^T$. $Q^T S \neq I$ in general. This is because we are using orthogonal left inverse to find Q_i^T . More on this later as we will dig into oblique projection, dual Gram-Schmidt process later. For now, we can just calculate the Q^T as S^{-1} using RREF.

3 Core idea of testing if A is diagonalizable

If A is diagonalizable depends on if each eigenspace's geometrical dimension is the same as the eigenvalue's algebraic multiplicity.

But how do we know once we get eigenvalue from power method?

Let us look at the more general Canonical Jordan decomposition of a linear transformation A . Our diagonalizable matrix is just a special form of CJM. From CJM point of view, the input $\mathbb{R}^n$ is direct sum of extended eigenspace:

$$\mathbb{R}^n = T_1 \oplus T_2 \oplus \cdots \oplus T_k$$

T_i here is extended eigenspace.

So what is extended eigenspace T_λ ? The following graph illustrates well:

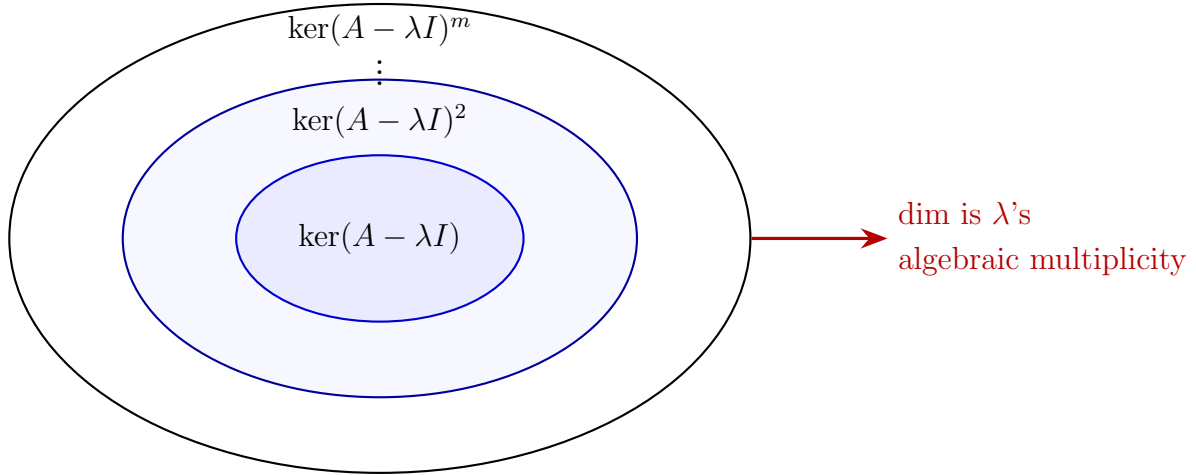


Figure 1: *
Nested Extended Eigenspaces (The multi-layer onion)

Obviously:

$$\ker(A - \lambda I) \subseteq \ker(A - \lambda I)^2 \subseteq \cdots \subseteq \ker(A - \lambda I)^m = \ker(A - \lambda I)^{m+1}$$

It is like a multi-layer onion. The innermost layer is $\ker(A - \lambda I)$ aka E_λ .

If we keep expanding, at one point $\ker(A - \lambda I)^m = \ker(A - \lambda I)^{m+1}$. Because the dim of A is limited, it must stop somewhere.

For diagonalizable A , all the extended eigenspace T is the same as E_λ . Put it another way:

$$\ker(A - \lambda I) = \ker(A - \lambda I)^2$$

This is the test we are looking for.